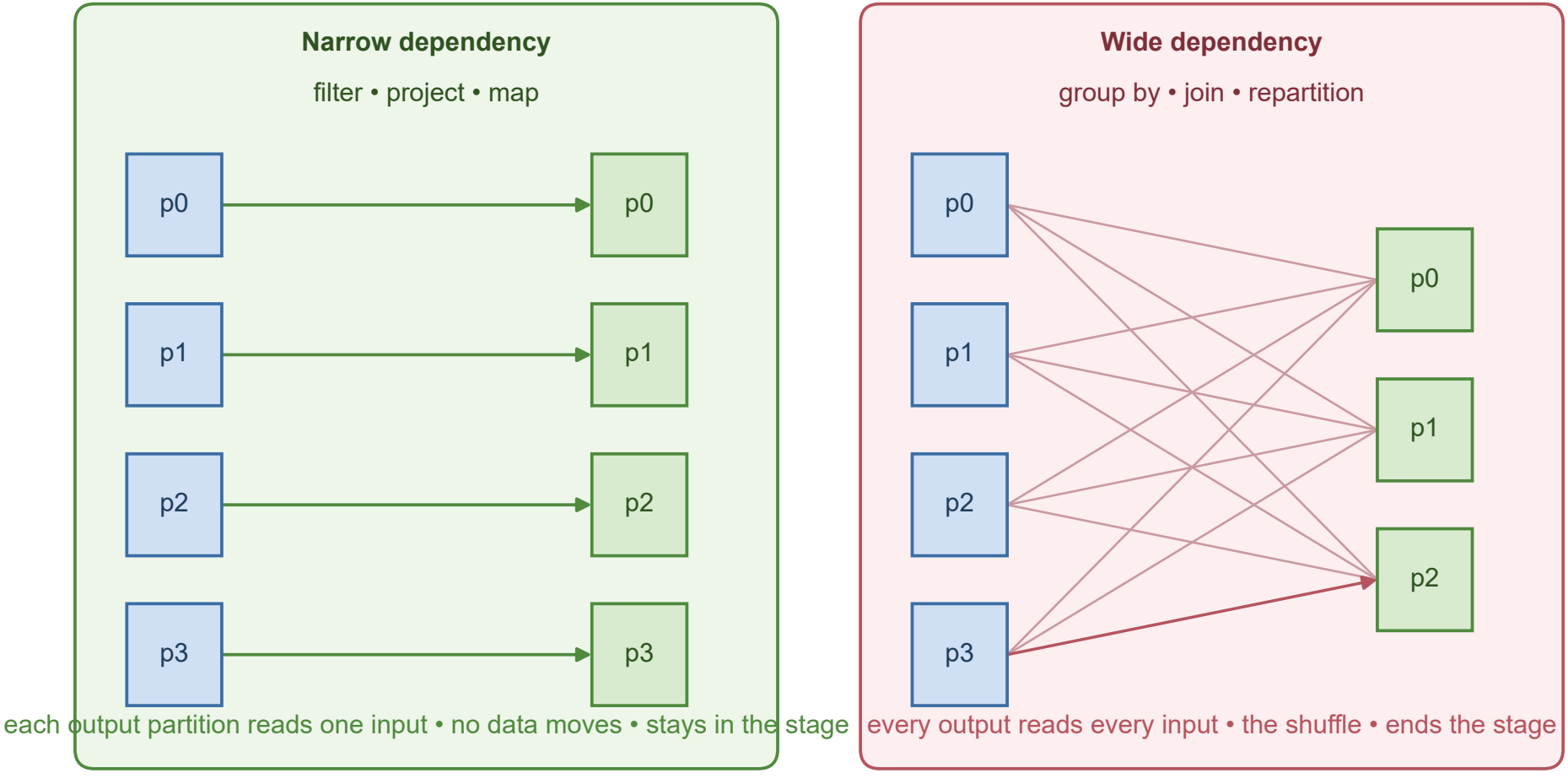
Narrow dependency
filter • project • map
p0
p1
p2
p3
p0
p1
p2
p3
each output partition reads one input • no data moves • stays in the stage
Wide dependency
group by • join • repartition
p0
p1
p2
p3
p0
p1
p2
every output reads every input • the shuffle • ends the stage